

Dynamics around Charged Black Holes in Entangled Relativity

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The two principles of General Relativity:

- General covariance
- Equivalence principle

Mach's principle: The metric tensor $g_{\mu\nu}$ should be **completely** determined by the energy-momentum tensor $T_{\mu\nu}$.

2015: Entangled Relativity

Conventions: $(-, +, +, +)$ metric signature and $c = \hbar = G = 4\pi\epsilon_0 = 1$ natural system of units.

Einstein-Hilbert action:

$$S[g, \Psi] = \int \left(\frac{R}{2\kappa} + \mathcal{L}_m \right) \sqrt{|g|} d^4x \quad (1)$$

Einstein field equations:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa T_{\mu\nu} \quad (2)$$

with $T_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\partial(\sqrt{|g|}\mathcal{L}_m)}{\partial g^{\mu\nu}}$.

Reissner-Nordström solution:

$$\begin{aligned} ds^2 = & - \left(1 - \frac{r_+}{r}\right) \left(1 - \frac{r_-}{r}\right) dt^2 \\ & + \left(1 - \frac{r_+}{r}\right)^{-1} \left(1 - \frac{r_-}{r}\right)^{-1} dr^2 \\ & + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2) \end{aligned} \quad (3)$$

Horizons:

$$r_{\pm} = M \pm \sqrt{M^2 - Q^2} \quad (4)$$

The solution exists for $|Q|/M \leq 1$.

Action of Entangled Relativity:

$$S[g, \Psi] = \int \frac{\mathcal{L}_m^2}{R} \sqrt{|g|} d^4x \quad (5)$$

Field equations:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = -\frac{R}{\mathcal{L}_m} T_{\mu\nu} + \frac{R^2}{\mathcal{L}_m^2} (\nabla_\mu \nabla_\nu - g_{\mu\nu} \square) \frac{\mathcal{L}_m^2}{R^2} \quad (6)$$

Local non-conservation of energy-momentum:

$$\nabla_\mu T^{\mu\nu} \neq 0 \quad (7)$$

Scalar-Tensor form of Entangled Relativity

Non-linear $f(R)$ theories admit an additional degree of freedom. Classically equivalent form (for $\mathcal{L}_m \neq 0$):

$$S[g, \phi, \Psi] = \int \left(\frac{\phi R}{2} + \sqrt{\phi} \mathcal{L}_m \right) \sqrt{|g|} d^4x \quad (8)$$

or with $\phi = \kappa^{-2}$:

$$S[g, \kappa, \Psi] = \int \frac{1}{\kappa} \left(\frac{R}{2\kappa} + \mathcal{L}_m \right) \sqrt{|g|} d^4x \quad (9)$$

Field equation:

$$3\kappa^2 \square \kappa^{-2} = \kappa (T - \mathcal{L}_m) \quad (10)$$

Charged Black Holes in Entangled Relativity

Charged black hole solution of ER:

$$\begin{aligned} ds^2 = & - \left(1 - \frac{r_+}{r}\right) \left(1 - \frac{r_-}{r}\right)^{15/13} dt^2 \\ & + \left(1 - \frac{r_+}{r}\right)^{-1} \left(1 - \frac{r_-}{r}\right)^{-7/13} dr^2 \\ & + r^2 \left(1 - \frac{r_-}{r}\right)^{6/13} (d\theta^2 + \sin^2 \theta d\varphi^2) \end{aligned} \quad (11)$$

Scalar Field:

$$\phi = \left(1 - \frac{r_-}{r}\right)^{-4/13} \quad (12)$$

Einstein frame obtained by:

$$g_{\mu\nu} \mapsto \tilde{g}_{\mu\nu} = \phi g_{\mu\nu} \quad (13)$$

New solution:

$$\begin{aligned} d\tilde{s}^2 = & - \left(1 - \frac{r_+}{r}\right) \left(1 - \frac{r_-}{r}\right)^{11/13} dt^2 \\ & + \left(1 - \frac{r_+}{r}\right)^{-1} \left(1 - \frac{r_-}{r}\right)^{-11/13} dr^2 \\ & + r^2 \left(1 - \frac{r_-}{r}\right)^{2/13} (d\theta^2 + \sin^2 \theta d\varphi^2) \end{aligned} \quad (14)$$

Einstein and Entangled Frames

Entangled frame

$$g_{\mu\nu}$$

Massive particles do not follow geodesics

Einstein frame

$$\tilde{g}_{\mu\nu} = \phi g_{\mu\nu}$$

Particles follow geodesics

Schematically: " $\tilde{g} \equiv g + \phi$ "

Domain of Validity of the Solution

Horizons:

$$r_- = \frac{13}{11} \left(M - \sqrt{M^2 - \frac{11}{12} Q^2} \right) \quad (15)$$

$$r_+ = M + \sqrt{M^2 - \frac{11}{12} Q^2} \quad (16)$$

Two conditions:

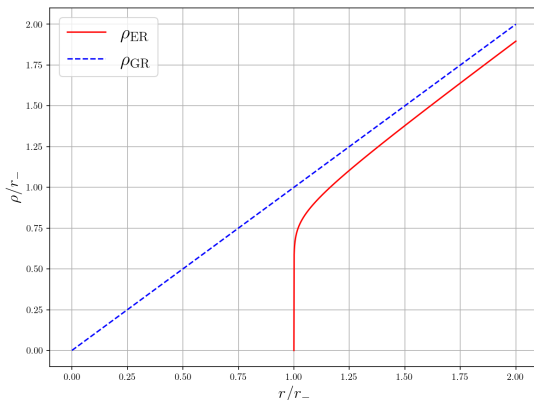
- Real-valued radii: $|Q|/M \leq \sqrt{12/11}$
- Inner horizon lower than outer horizon: $r_- \leq r_+$

Solution exists for $|Q|/M \leq \sqrt{143/132}$.

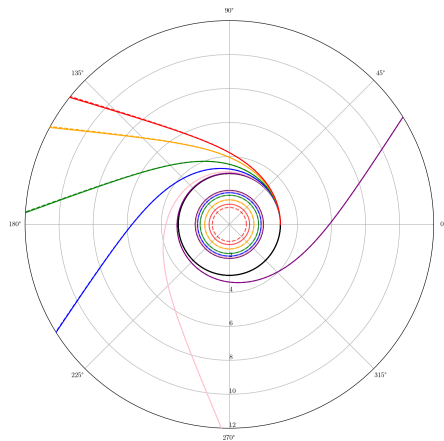
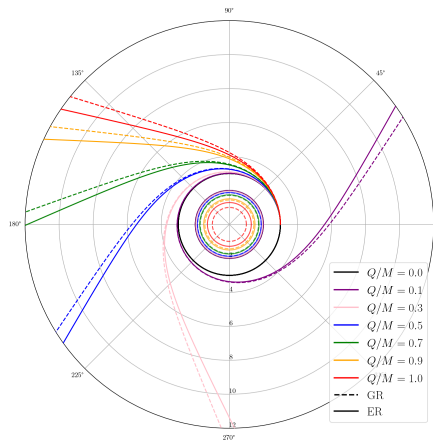
Areal Radius

Areal radius ρ : $A = 4\pi\rho^2$

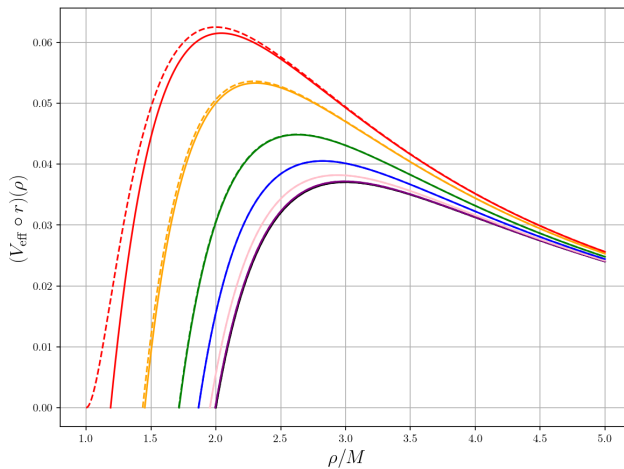
$$\rho_{\text{GR}} = r_{\text{GR}}, \quad \rho_{\text{ER}} = r_{\text{ER}} \left(1 - \frac{r_-}{r_{\text{ER}}}\right)^{1/13} \quad (17)$$



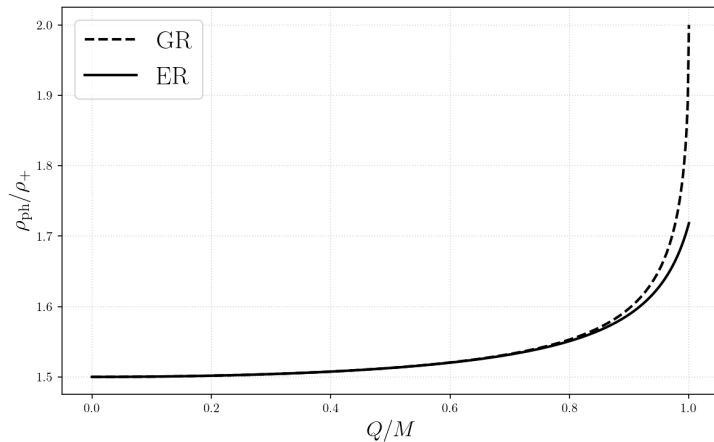
Null Geodesics Analysis



Null Geodesics Analysis - Effective Potential



Null Geodesics Analysis - Photon Sphere



Timelike Geodesics Analysis - ISCO Radius

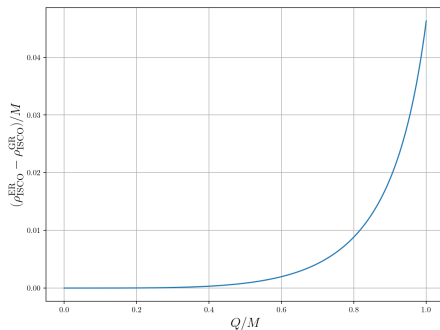
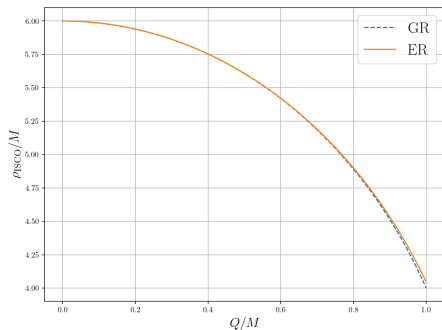
Given in GR by:

$$Mr^3 - 6M^2r^2 + 9MQ^2r - 4Q^4 = 0 \quad (18)$$

Given in ER by:

$$\begin{aligned} & (2197r_+ + 1859r_-)r^4 \\ & - (6591r_+^2 + 13351r_+r_- + 6578r_-^2)r^3 \\ & + (24336r_+^2r_- + 25740r_+r_-^2 + 4620r_-^3)r^2 \\ & - (31668r_+^2r_-^2 + 14388r_+r_-^3)r \\ & + 13824r_+^2r_-^3 = 0, \end{aligned} \quad (19)$$

Timelike Geodesics Analysis - ISCO Radius



Timelike Geodesics Analysis - Kepler's Third Law

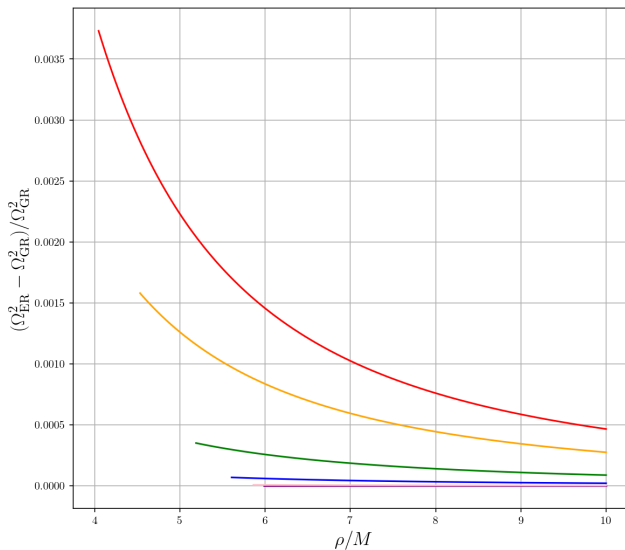
Given in GR by:

$$\Omega_{\text{GR}}^2 = \frac{M}{r^3} - \frac{Q^2}{r^4} \quad (20)$$

Given in ER by:

$$\Omega_{\text{ER}}^2 = \frac{\left(1 - \frac{r_-}{r}\right)^{9/13}}{1 - \frac{12}{13} \frac{r_-}{r}} \Omega_{\text{GR}}^2 \quad (21)$$

Timelike Geodesics Analysis - Kepler's Third Law



Black Hole Phenomenology - Generation of Mock Data

True parameters: $M_{\text{true}}, q_{\text{true}} = |Q_{\text{true}}|/M_{\text{true}}$.

N randomly generated areal radii:

$$1.01 \max(\rho_{\text{ISCO}}^{\text{GR}}, \rho_{\text{ISCO}}^{\text{ER}}) < \rho_i < 15M_{\text{true}} \quad (22)$$

Generating mock data via ER:

$$\Omega_{i,\text{true}} = \Omega_{\text{ER}}(\rho_i; M_{\text{true}}, q_{\text{true}}) \quad (23)$$

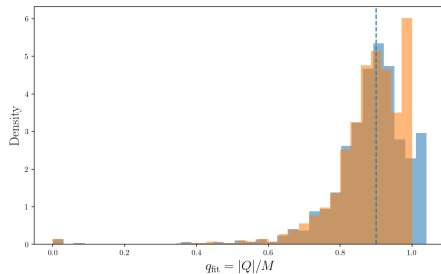
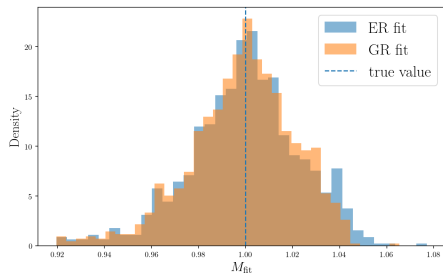
Adding gaussian noise:

$$\Omega_{i,\text{obs}} = \Omega_{i,\text{true}} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_{\Omega_i}^2) \quad (24)$$

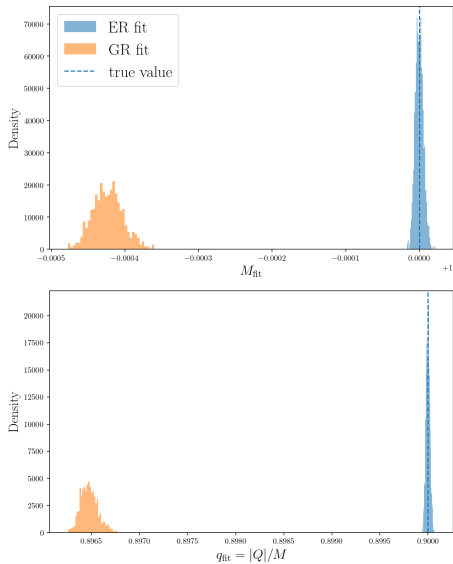
Fitting by minimizing:

$$\chi^2(M, q) = \sum_{i=1}^N \frac{[\Omega_{i,\text{obs}} - \Omega(\rho_i; M, q)]^2}{\sigma_{\Omega_i}^2}. \quad (25)$$

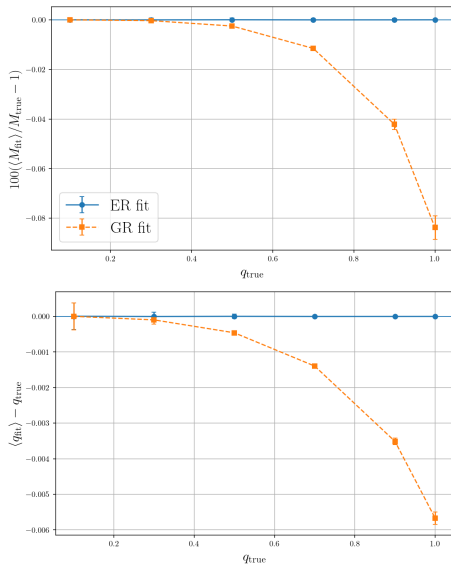
Black Hole Phenomenology - $N = 8$, $\sigma_\Omega/\Omega = 1\%$



Black Hole Phenomenology - $N = 100$, $\sigma_\Omega/\Omega = 10^{-5}$



Black Hole Phenomenology - $N = 100$, $\sigma_\Omega/\Omega = 10^{-5}$



Results summary:

- ER allows for a broader range of charge-to-mass ratio.
- The causal structure of an ER charged black hole spacetime is not the same as a Reissner-Nordström spacetime.
- ER predicts a smaller photon sphere.
- ER predicts a larger ISCO radius.
- ER predicts a larger rotation velocity in circular orbits.
- GR tends to underestimate the properties of a black hole.

Main result: Even for extremely charged black holes, the two theories are nearly impossible to distinguish.

Thank you for listening.

Thanking Olivier Minazzoli and:

